SECURITY ASSESSMENT

Grand Marina Hotel Water Monitoring System

Prepared for: Marcus Webb, General Manager
Prepared by: Emad Samara, Junior Security Analyst
Date: September 19, 2026

EXECUTIVE SUMMARY

I recommend turning on encryption for all 50 sensors in Grand Marina's water monitoring system as soon as possible. Encryption (the technical name is TLS) scrambles information so that only authorized systems can read it. Right now, the water sensors send their readings, water pressure and flow rate, across the network in plain, readable form. In our test, a free, standard program and one line of typing were enough to read every message a sensor sent.

After we turned encryption on, the same attempt showed nothing. Encryption added about 0.14% to the time a reading takes to arrive, roughly 4,000 times shorter than a blink of an eye, and the system handled double our emergency-level workload with no errors. For a hotel with 2,000 guests, that means better protection with no change anyone would notice. One condition matters: the system's ID check must stay switched on. When we switched it off in our test, the connection was accepted without any questions.

SECURITY RISKS ADDRESSED

Before Encryption

Today, every reading from every water sensor travels across the network where anyone who looks can read it. It is like reading meter numbers aloud in a crowded lobby instead of passing a sealed note. In our test, a listener captured all five messages our test sensor sent. Each one showed the sensor's ID, the water pressure, and the flow rate. No special access was needed. For a hotel with 2,000 guests depending on working water every day, that is an operational and reputational risk, not just a technical one.

What this means for Grand Marina:

- Anyone who finds a way onto the network, such as a contractor or a former employee, could silently watch the water system in real time.
- Flow readings rise and fall with water use, so they could hint at when the hotel is busy or quiet.
- Without a check on who is connecting, someone could potentially send false readings, hiding a real leak or triggering a false alarm. With 2,000 guests in the building, a missed leak or a needless emergency response could mean disrupted rooms, guest complaints, and costly repairs. We did not test this directly, but it is a recognized risk of unprotected systems.
- **Hotels are real targets.** In September 2023, a cyberattack on MGM Resorts disrupted digital room keys and reservations. In March 2024, an attack on Omni Hotels & Resorts forced systems offline, including reservations, payments, and digital room access. That same month, researchers found a flaw in Saflok keycard locks used on more than three million doors in 131 countries: someone holding an ordinary keycard could forge cards that open doors across the property (no real-world use was known at the time). To be fair, none of these was publicly traced to unprotected sensor data (MGM's attackers reportedly

got in through social engineering). They do show how far one incident can reach into guest safety and daily operations.

- **The cost is real.** IBM's 2025 study, based on Ponemon Institute research, puts the average hospitality data breach at $4.03 million, up from $3.82 million the year before. Across industries, about one in three breaches studied (32%) ended in a regulatory fine. If Grand Marina handles data on EU guests, GDPR fines can reach €20 million or 4% of worldwide annual turnover, whichever is higher. That is a ceiling, not a typical fine: as of May 2026, regulators had issued 91 GDPR fines to hospitality businesses in 15 countries, totaling about €22.7 million. Water readings are not guest records, but regulators look at a hotel's overall security practices, and leaving a system unprotected weakens the hotel's position.

After Encryption

- All readings are scrambled in transit. In our test, the listener was refused and saw nothing readable.
- The system blocks a device that presents the wrong ID, as long as the ID check is switched on.
- Grand Marina gets a tested, documented answer if insurers, partners, or regulators ask how the building's systems are protected.

What encryption will not do: it does not stop fake emails or phone calls (how MGM's attackers reportedly got in), weak passwords, or outdated device software. It is one important layer, not the whole wall. Fire alarms and emergency shutoffs should also stay on their own dedicated local systems rather than depend on this network.

PERFORMANCE IMPACT

The key question: does adding security slow down water monitoring?

What we measured	Without encryption	With encryption	Verdict
Average time for a reading to arrive	50.97 ms	51.04 ms	No noticeable difference
Fastest reading	0.49 ms	0.63 ms	No noticeable difference
Slowest reading	100.99 ms	101.34 ms	No noticeable difference

ms = one thousandth of a second. Each result is from 50 test messages.

The bottom line: encryption added 0.07 thousandths of a second to the average reading, an increase of 0.14%. A blink takes roughly 300 thousandths of a second, so the delay is about 4,000 times shorter than a blink. The average reading arrives in about 51 thousandths of a second either way. Building-industry guidance suggests under one to two seconds for non-critical alarms and equipment notices, and one to ten seconds for routine readings. These are engineering guidelines, not legal standards.

Real devices can show a larger gap than our lab did. A 2026 study of a small sensor chip found encryption raised average delivery time by about 14%, from 45 to 52 thousandths of a second. That is still far too small for staff to notice. The extra time shows up mostly when a sensor first connects, so sensors should stay connected rather than reconnect for every reading.

Capacity: we tested up to 100 messages per second, double the 50 per second we treated as the emergency level. With encryption on, the system delivered 96.8 per second over 3,000 messages with zero errors.

TESTING EVIDENCE

We ran four tests. They used a test sensor in our lab setup, not Grand Marina's live system, so real-world numbers may differ slightly. We recommend repeating the key checks after rollout.

1. Can someone spy on our sensor data? (Eavesdropping Test)

Without encryption: the listener saw all 5 messages, with sensor ID, water pressure, and flow rate in plain text. **With encryption:** the same attempt failed with a connection error and showed no messages.

2. What happens when the ID check is wrong or switched off? (Certificate Test)

Certificates are digital ID badges that let the system confirm who it is talking to.

Scenario	Result	What happened
Correct ID badge	Connected	System works normally
Wrong ID badge	Blocked	Certificate verification error
ID check switched off	Connected	Accepted without checking who was on the other end

What this tells us: the ID check does its job, but only when it is on. It must be locked on, with no setting that lets a sensor skip it.

3. Does encryption slow things down? (Speed Test)

Average delivery time went from 50.97 to 51.04 thousandths of a second, a 0.14% increase. That is negligible for water monitoring.

4. Can it handle a crisis? (Stress Test)

Load (messages per second)	Without encryption	With encryption
10 (normal)	Success	Success
25 (moderate)	Success	Success
50 (emergency)	Success	Success
100 (beyond what we expect to need)	Success	Success

Even at 100 per second, encryption caused no failures.

RECOMMENDATION

Turn on encryption for all 50 water sensors and the central monitoring system, and make the ID check mandatory.

- Right now, anyone on the network can read the water system's data, and 2,000 guests rely on that system.
- Encryption removes that exposure at a cost of 0.14% in speed, which no one at the hotel will notice.
- The system handled double our emergency load, so capacity is not a concern.

- The ID check must stay on, or the protection can be bypassed.

The cost of doing nothing is also real: hospitality breaches averaged $4.03 million in IBM's 2025 study. The protection is tested. The remaining step is to switch it on and keep it on.

NEXT STEPS

1. **Turn on security for all 50 sensors (Weeks 1–2).** Once this is done, every reading from every sensor is scrambled, and any sensor that tries to send readable readings is turned away.
2. **Lock down who can connect (Weeks 3–4).** Give each of the 50 sensors its own ID badge, make the badge check mandatory, and repeat the wrong-badge test to confirm impostors are turned away.
3. **Train the team and set up alerts (Weeks 4–5).** Brief staff on what to watch for and what to do if a sensor goes offline. Set up automatic alerts so the team knows right away.
4. **Document and separate (Weeks 5–6).** Record the tests for insurance and compliance files, and keep the water sensors separate from guest Wi-Fi and payment systems.

Sources

IBM / Ponemon Institute, Cost of a Data Breach Report 2025 (ibm.com)

CMS, GDPR Enforcement Tracker Report: Accommodation and Hospitality, May 2026 (cms.law); EDPB Guidelines 04/2022 on calculating GDPR fines (edpb.europa.eu)

SecurityWeek, Saflok lock vulnerability (securityweek.com); Canopy, MGM casino incident (gocanopy.com); Asimily, hospitality cyberattacks 2023–2025 (asimily.com)

Azhar, Azwar & Asroni (2026), MQTT vs. MQTT/TLS on ESP32-C3, JAIC (jurnal.polibatam.ac.id)